

Week 11: Testing and classroom delivery

Generative AI | MSc teaching notes and further reading | 2027

Learning target

Record a safe release decision with test and rollback evidence.

Context and teaching plan

A new policy-assistant version sometimes exposes a staff-only passage.

Contact time	Suggested use
Two hours: concepts	Case introduction, four ideas, worked example and misconception.
Two hours: practice	10 min recall, 20 min demo, 40 min guided practice, 25 min independent change, 15 min review, 10 min exit check.

Explain the key ideas

Release gate

Require evidence that inputs, permissions, output and failure handling meet a defined threshold.

Teacher question: Where does release gate appear in the weekly case, and what could go wrong if we omit it?

Filtered trace

Log enough to debug while omitting unnecessary personal data and credentials.

Teacher question: Where does filtered trace appear in the weekly case, and what could go wrong if we omit it?

Rollback

Restore a previously tested configuration if a new version fails.

Teacher question: Where does rollback appear in the weekly case, and what could go wrong if we omit it?

Cost per success

All run costs including failed attempts divided by successful tasks; undefined at zero success.

Teacher question: Where does cost per success appear in the weekly case, and what could go wrong if we omit it?

Worked example and answer guide

- Input: Version 1 blocks staff policy for student role.
- Change or question: Version 2 returns a staff paragraph.
- Supported observation: Release is blocked and version 1 remains active.

Reasoning: state the given evidence first, then the calculation or rule, then the conclusion. If a value is absent, say what cannot be concluded. Keep any simulated mechanism distinct from a trained model or deployed service.

Teach the week as a story

The failed classroom release: Version 2 answers faster but exposes one staff-only policy to a student. Would you release it?

Ask for a quick prediction before each technical explanation. Return to the prediction after the example and ask what evidence changed the answer. The analogies below are memory aids, and each has a stated limit.

Release gate: analogy, mechanism and check

Familiar picture: An aircraft checklist must pass every required item before departure.

Familiar part	Technical meaning
Candidate version	aircraft
Required tests	checklist
Release	departure decision

Where this analogy breaks: A checklist reduces known risk; it cannot prove zero future failures.

Actual idea: Require evidence that inputs, permissions, output and failure handling meet a defined threshold.

Work it aloud: Candidate v2 fails staff access, so the release decision is “block” even if other scores improve.

Ask before revealing: Can a faster answer override a failed access gate?

Supported response: No. The required access test must pass.

Look for this misconception: Most tests passing is enough to release.

Filtered trace: analogy, mechanism and check

Familiar picture: A shipping receipt records what happened without copying the private parcel.

Familiar part	Technical meaning
Event	trace record

Private text	removed content
Receipt	filtered log

Where this analogy breaks: Aggressive filtering can hide useful debug details; keep safe IDs and event type.

Actual idea: Log enough to debug while omitting unnecessary personal data and credentials.

Work it aloud: Log request ID, role, policy ID and outcome; omit student question text and secret key.

Ask before revealing: Which fields help investigate without exposing private text?

Supported response: Safe identifiers, role, source ID and decision.

Look for this misconception: A useful trace must store the full private prompt.

Rollback: analogy, mechanism and check

Familiar picture: A theater switches back to the last working lighting cue.

Familiar part	Technical meaning
New cue	candidate version
Old cue	tested configuration
Switch	rollback

Where this analogy breaks: Rollback needs tested artifacts and compatible data; rehearsal matters.

Actual idea: Restore a previously tested configuration if a new version fails.

Work it aloud: After v2 leaks P5, restore v1 with its policy index and confirm the student test passes.

Ask before revealing: What must be checked after restoring v1?

Supported response: Re-run the critical access and answer checks.

Look for this misconception: Rollback means changing only the visible version label.

Cost per success: analogy, mechanism and check

Familiar picture: A restaurant compares the full meal bill with meals actually served.

Familiar part	Technical meaning
Full bill	total run cost
Served meals	successful answers
Unit cost	cost per success

Where this analogy breaks: A very small sample can make the estimate noisy; state the denominator.

Actual idea: All run costs including failed attempts divided by successful tasks; undefined at zero success.

Work it aloud: If ten runs cost €0.20 and eight meet criteria, cost per success = €0.025.

Ask before revealing: Why divide by eight rather than ten?

Supported response: Only eight runs delivered the defined successful outcome.

Look for this misconception: Cheapest run is automatically best delivery.

New situation for discussion

The new version passes answer checks but fails the missing-source test. What does the release gate decide?

Memory cue: A required access test blocks release; keep safe traces and a tested rollback.

Guided laboratory

Run a local release-gate simulation and rollback.

1. Run approved and unauthorized access cases
2. Simulate a missing source and bad input
3. Filter a trace before sharing
4. Reject a bad candidate version and restore the prior config

Individual change: Flip the candidate role check and verify rollback happens. Expected evidence: Filtered trace and classroom release checklist.

Checks and common error

Measure: Passed release gates / required gates; cost per successful task. Record the input, setting, source version, result and the relevant numerator and denominator.

Plausible incorrect claim: A high answer accuracy can compensate for an access-control failure.

Correction: Access is a release gate. Report accuracy and security outcomes separately.

If learners disagree, reproduce the same case and compare the underlying source or expected answer. Record unresolved cases for instructor review.

Further discussion

- How could the conclusion change if the source or input changes?
- Which part of the result is observed and which part is an assumption?
- What simple test would detect the failure described above?

References and optional extension

Core readings are official vendor documentation or original research. Read the relevant section; an account or paid API is unnecessary for this notebook. Page details may change before 2027.

OpenAI, Agent safety. <https://developers.openai.com/api/docs/guides/agent-builder-safety>

OpenAI, Evaluation best practices. <https://developers.openai.com/api/docs/guides/evaluation-best-practices>

NIST, Generative AI Profile. <https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf>

Optional extension: replace one classroom proxy with a locally approved model or service, keep the original baseline, repeat the same checks, and state the additional privacy and cost assumptions.

Curriculum scope: the supplied 12-week teaching sequence, recorded in docs/SYLLABUS_ALIGNMENT.md. The original module descriptor is retained in sources/originals/07-Descriptor-Generative-AI-new-.docx. Supporting project notes: Deep-research.txt, RAG.txt, AI-Agents.txt, MCP-&-Vector-Database.txt, Multi-Agent-AI-System.txt, Eval-&-Fine-tune-&-Loop-Engineering.txt. These are planning sources, not empirical performance evidence.

Student revision and exam preparation

Outcomes: LO4, LO5 | 2 hours concepts + 2 hours guided practice

The idea in plain language

A classroom release has a versioned configuration, tested behavior and a rollback path. Test failures by actually supplying bad inputs or removing evidence. Keep only necessary non-sensitive trace fields. Calculate cost using all attempts, including failures and retries.

A worked example

If total simulated spend is 4 units and eight tasks succeed, cost per success is 0.50 units. If no tasks succeed, report undefined. If a candidate version exposes a staff source to a student, keep the previous version even if average answer accuracy improves.

Practice before reading the answers

1. What happens to cost per success when successes equal zero? (4 suggested practice marks)
2. Why test a missing source before release? (4 suggested practice marks)
3. What should a privacy-filtered trace retain? (4 suggested practice marks)

Draw the main diagram from memory, explain each arrow, then check it against the slides. Answer the three questions without notes before reading the model answers.

Model answers and marking guidance

1. There is no finite value. Report undefined and investigate the failed workload.
2. To verify the application abstains or recovers rather than inventing an answer.
3. A request ID, source IDs, version, outcome and timing as needed; omit credentials and unnecessary personal content.

For each four-mark practice answer: two marks for the correct idea, one for a relevant example or calculation, one for an appropriate explanation or limitation. These suggested practice marks do not change the module assessment scheme.

Official reading and video support

M3 Microsoft Learn: Building a RAG-powered AI chat app with Python and VS Code

<https://learn.microsoft.com/en-us/shows/vs-code-day-2024/building-a-rag-powered-ai-chat-app-with-python-and-vs-code>

Study focus: Optional system walkthrough. Older SDK interfaces are not the core lab dependency.

O1 OpenAI: Evaluation best practices

<https://developers.openai.com/api/docs/guides/evaluation-best-practices>

Study focus: Task-specific testing, repeated outputs and separate evaluation cases.

O2 OpenAI: Safety in building agents

<https://developers.openai.com/api/docs/guides/agent-builder-safety>

Study focus: Tool approvals, untrusted input and controls.

Selected source concepts inform the lesson. Examples, figures, questions and solutions are original classroom material; no exam questions are claimed to be official.